Assignment 2

Part 1: Polynomial Basis Function

A)

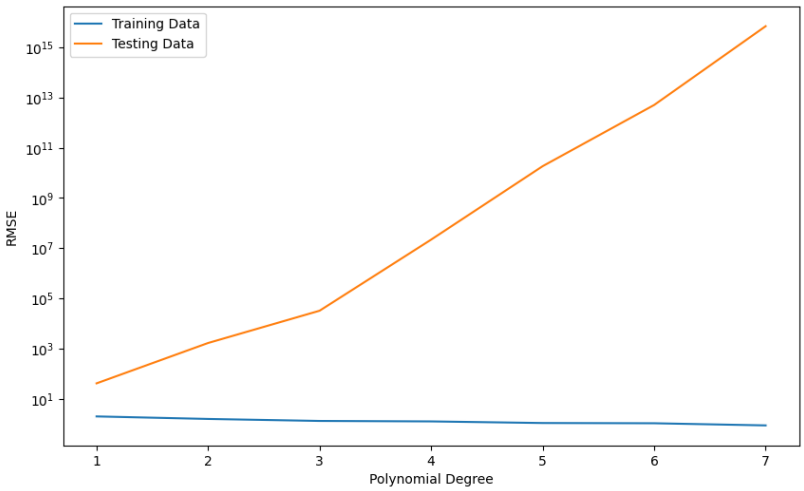


Figure 1: Root Mean Squared Error for Polynomial Regression.

As the polynomial degree increases the training data RMSE approaches 0 as testing RMSE deviates and progressively increases. At the high polynomial degrees the model is overfitting to the training data resulting in very low RMSE error on training data and inversely high RMSE on the validation dataset.

B)

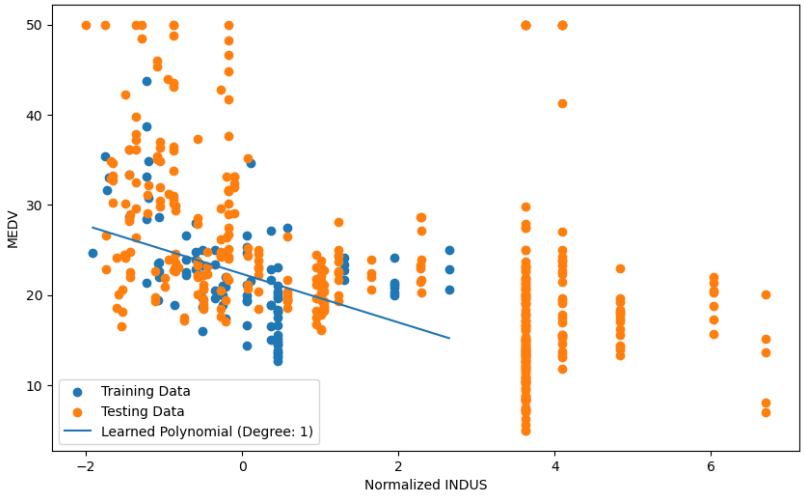
Table 1: weights for polynomial regression using a degree 1 polynomial

Stat	**Weights**
BIAS	22.344000
CRIM	-1.548640
ZN	0.152050
INDUS	-0.596136
CHAS	-0.000000
NOX	1.297793
RM	4.222276
AGE	-2.195168
DIS	0.019016
RAD	-0.286457
TAX	-0.227556
PTRATIO	0.242083
B	0.148338
LSTAT	0.075028

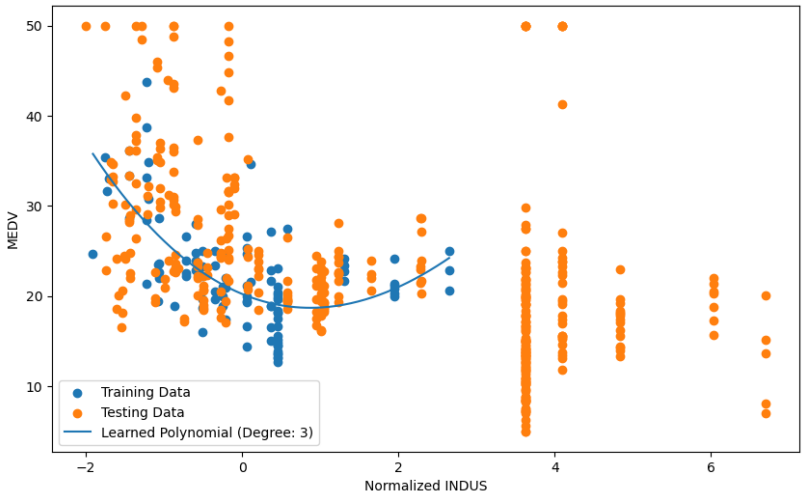
What value is chosen for w5, the weight on the 6th feature (average number of rooms per dwelling)? What value is chosen for the weight on the 8th feature (weighted distance to five Boston employment centers)? (Don't forget the bias weight and the normalizations.) Do these 2 weights seem reasonable?

The RM and DIS weights are 4.22 and 0.019 respectively. For the RM values a higher number of rooms per dwelling would be very likely to increase the value of a given home therefore a positive weight for this metric is reasonable. For the DIS value, a home closer to an employment center would generally be valued at a higher price due to the convenience of the location and the positive value for this metric is reasonable. The choice of a much higher weight for RM metric compared the DIS metric also makes sense as the number of rooms in a home will likely have a much higher effect on the home price than the relative location with respect to an employment center.

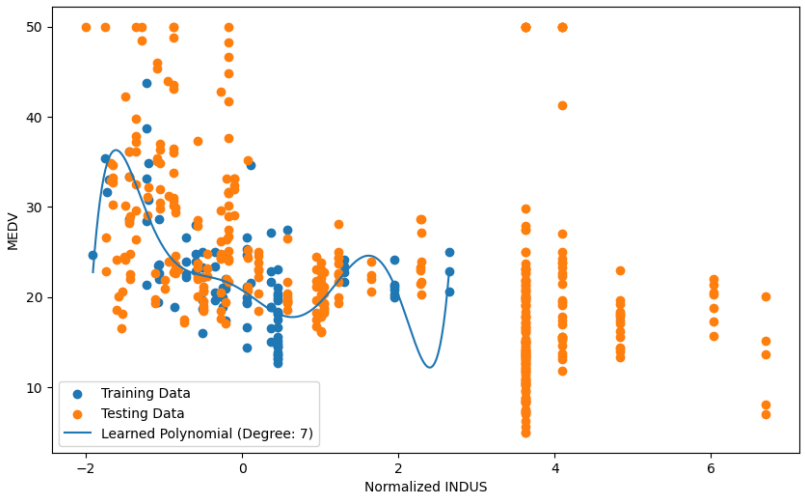
C)



(a) 1D Polynomial Regression



(b) 3D Polynomial Regression



(c) 7D Polynomial Regression

Figure 2: INDUS Polynomial Regression

As the polynomial degree increases the trend line for the learned polynomial display more local minima and maxima, fitting better to the training dataset as the degree is increased. The overfitting effect discussed in section 1a can be clearly observed in the plots above, both the mid and high order polynomial trend lines, while objectively modeling the training dataset better than the low order polynomial, do not accurately capture the linear negative trend of the testing dataset.

D)

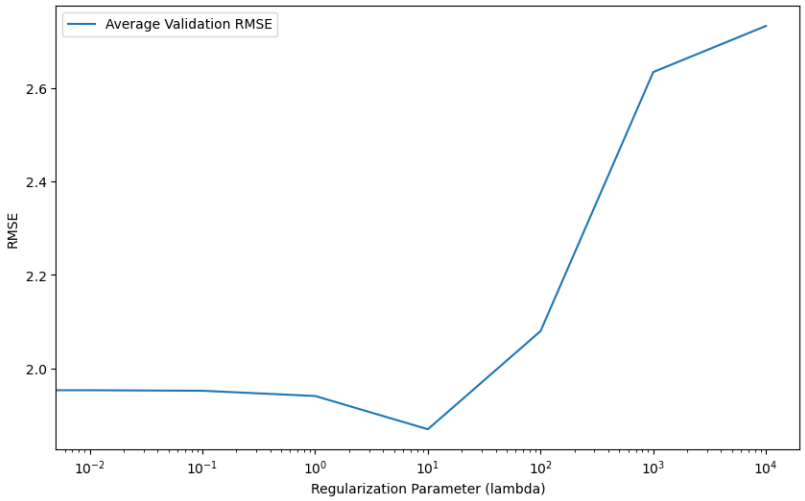


Figure 3: Root Mean Squared Error vs Regularizing Constant

Given the plot above, I would choose a lambda value of 0.

Part 2: Gaussian Basis Function

E)

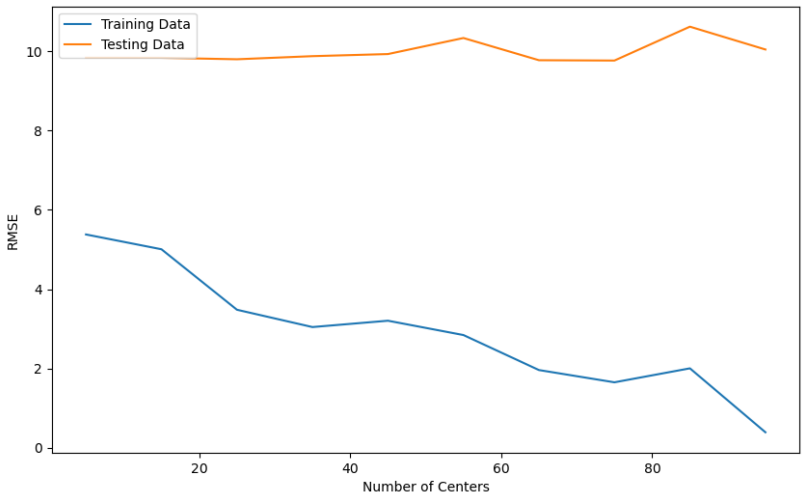


Figure 4: Root Mean Squared error vs Gaussian Basis Functions

As the number of gaussian basis points increases, the training data RMSE decreases similar to 1a while the testing data remains at a relative constant value. The same overfitting present in 1a occurs here however unlike the high order weights the additional centers that overfit to the training set have no effect on the training data (Note this doesn't mean the additional centers are improving accuracy, overfitting still provides no benefit when using the model on new data and just increases computation)

F)

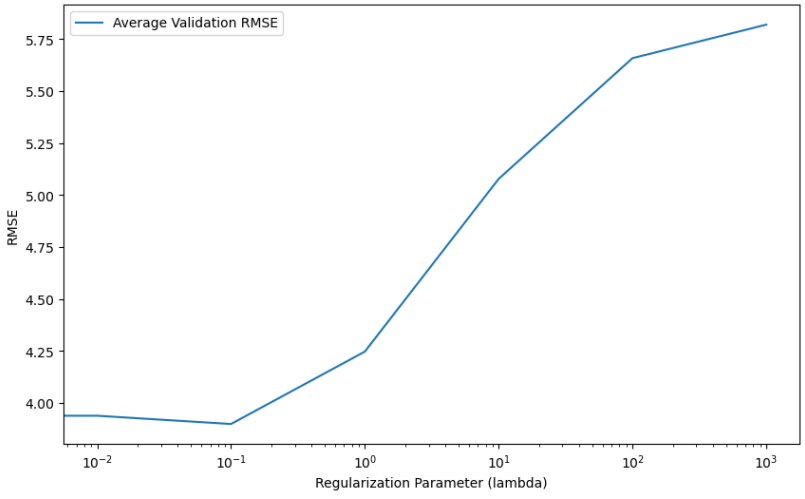


Figure 5: Root Mean Squared Error vs Regularizing Constant

Given the plot above, I would choose a lambda value of 0.1 .